

Code a Dance Party!

Choreographing with Scratch Jr 

 MIXING CODING & DANCE MOVES! 

Hey Magical Code Dancers! 

Have you ever gone to a festive wedding, danced at a school Annual Day, or celebrated a fun festival like Diwali or Navratri? The incredible rhythm, the stepping, and the spinning you do are just like writing code for a computer! In this lesson, we will become master dance choreographers and use simple coding blocks to create an epic, animated dance party screen!

Target Grade Level: Kindergarten to Grade 2 (Ages 5-8) ✨

Platform: Scratch Jr App (Tablet / Mobile Interface)

Designed & Formatted Professionally by Planrs Academy Curriculum Labs Division.

1. The Big Stage Presentation Hook

Let's remember a super exciting school day. Imagine the stage lights are shining brightly, music is booming loudly from the speakers, and you are wearing a beautiful, glittering costume for your ****Annual Day Dance Performance****! Your heart is beating fast with excitement.

WHAT IS A CHOREOGRAPHER?

Before you went out onto that big stage, a dance teacher stood in front of you for weeks showing you exactly when to step left, when to clap your hands, and when to spin around. That teacher is a ****Choreographer****! Today, you are going to be the choreographer, but instead of training humans, you will code cute digital characters to dance perfectly to your beats!

The Little Coder Goals

- **Dance Steps = Code Blocks:** Discover how a real-life dance routine is identical to a computer program sequence.
- **Rhythm Adjustments:** Learn how changing numbers makes characters slide fast or hop slowly.
- **Looping Grooves:** Use repeating patterns so your characters keep grooving without stopping!

TEACHER VERBATIM SCRIPT: THE HOOK

Teacher (clapping to a 1-2-3 beat): "Listen to this rhythm, code dancers! Clap, clap, wave! Clap, clap, wave! That pattern is exactly how a computer thinks. Let's step inside our Scratch Jr playground and build our very own animated stage show!"

2. Transforming Big Words into Kid Concepts

Sometimes, programmers use giant, scary-sounding words that make coding sound difficult. But guess what? You already know exactly what they mean because you use them on the playground every day! Let's translate them together:

OUR SECRET CODING TRANSLATION CARD

Big Scary Word	What it Actually Means	Fun Dance Example
Linear Script	Doing steps in a straight line, one after another.	First you bow, then you jump, then you smile.
Sequence	The perfect order of your steps. Changing the order ruins the style!	Putting your shoes on *before* you tie the laces.
Parameter	A settings knob or number input that changes how big or fast an action is.	Taking 5 small hops instead of just 1 giant leap!
Loop Variable	A magic box that tells a step to repeat over and over again.	Clapping your hands 4 times continuously to match a drum.

By pairing these abstract technological ideas directly with physical dance habits, early learners retain computational structures effortlessly without any academic stress!

3. Warm-Up Game: The Human Code Machine!

Before we touch our screens or open tablets, let's stand up and turn our entire classroom into a live, physical computer network! This high-energy game prepares our brains to think like step-by-step programmers.

HOW TO PLAY THE GAME:

1. **Form Code Duos:** Split up into pairs with a partner. **Student A** is the *Software Programmer*, and **Student B** is the *Mechanical Robot Sprite*.
2. **Hold Up the Code Flashcards:** The Programmer holds up large, colorful cards that mimic programming commands, such as:  ,  , or  .
3. **Strict Robot Rules:** The Robot Sprite must perform these physical actions exactly as written, working linearly from start to finish without making any unwritten changes!

LIVE PLAY-LOOP SCRIPT

Teacher: "Programmer, hold up the [HOP TWICE] card! Robot, execute command now!... Splendid! Now look what happens if we place the [REPEAT 3 TIMES] block around it. How many times must our robot hop now?"

Students: "Six times total!" 🤖

4. Dance Style 1: The High-Energy Bhangra! 🌾

Our first animated dance style comes from the beautiful, sunny green fields of Punjab. It is called **Bhangra**! It is a super high-energy celebration dance filled with giant smiles, huge hops, and arms pointing high up to the sky.

THE BHANGRA PHYSICAL STEP GUIDE:

Stand up straight on your feet. Raise both of your hands up in the air above your shoulders. Now, bounce your right knee up high while pulsing your hands up to the sky, then switch to your left knee! It feels like you are bouncing on a fun trampoline.

HOW WE BUILD BHANGRA WITH SCRATCH JR BLOCKS:

To make our digital character perform this festive Bhangra groove, we must combine motion and size blocks into a perfect sequential train:

Scratch Jr Block	Block Group	What It Tells the Actor to Do
	Yellow Triggers	Signals the dance to start immediately when the game begins.
	Blue Motions	Commands our character to leap high into the air and land back down.
	Purple Looks	Changes the character's size parameters, making them bounce rhythmically toward the audience!

5. Dance Style 2: The Swirling Garba Spin! ○

Our next dance style travels to the festive streets of Gujarat. It is called **Garba**! This dance is performed during the huge Navratri festival where people gather in giant, beautiful circles to dance together under the stars.

THE GARBA PHYSICAL STEP GUIDE:

Imagine walking along a giant circular track painted on the floor. You take three graceful steps forward, clap your hands together tightly to make a loud sound, and then do a full, swirling circle spin right on your tiptoes!

TRANSLATING GARBA INTO COMPUTER CODE

How do we tell a computer to do a circular spin and a clap? We look for the rotational motion blocks and audio triggers in our block inventory menu:

🌀 THE GARBA LOGIC BLOCK FORMULA

- **The Walk Input:** Use the **Move Right** block and set the number parameter to ****4****.
- **The Festive Sound:** Attach the purple **Play Pop Sound** block to mimic the hand clap rhythm!
- **The Big Twirl:** Snap in the circular **Turn Clockwise** arrow and type in the number ****12**** to make a full, beautiful spin!

CHOREOGRAPHER DIALOGUE PROMPT

Teacher: "Watch my character on the slide! She steps, she claps, and then—look at that! —she spins around like a colorful top! That's the power of sequencing. Let's make sure our rotation number is high enough so she turns all the way around!"

6. Setting Up Your Digital Dance Stage

Now it's time to open your tablet, launch the Scratch Jr application, and set up our virtual theater production! Follow these setup steps carefully:

STEP 1: BUILDING THE THEATER BACKDROP

Click on the landscape picture icon at the very top header bar of your screen to browse the **Background Library**. For our grand dance party, look for the ****Stage**** backdrop (the one with the gorgeous red velvet curtains) or the festive ****City Park at Night**** setting.

STEP 2: CASTING YOUR DANCE TROUPE ACTORS

Go to your left-hand side character house. Delete the default cat by holding your finger down on it until it wiggles and shows an 'X'. Now, tap the big plus (+) button twice to add two custom performers:

YOUR ASSIGNED DANCE DUO

Performer 1: The Bhangra King 
Drag him to the left side of the stage.

Performer 2: The Garba Queen 
Drag her to the right side of the stage.

Creative Modification Track: If you can't find those specific human characters, you can customize any sprite! You can code a happy ****Elephant**** to do the Bhangra or a cute ****Frog**** to spin the Garba. Let your imagination run wild!

7. Assembling the Code Train for Bhangra

Click directly on your **Bhangra Performer** icon on the left side of your screen. Let's assemble his code puzzle blocks from left to right in the lower programming workspace canvas area.

THE BHANGRA CODE RECIPE GUIDE

Find these specific blocks and snap them together tightly so there are no loose spaces between them:

1. **Green Flag Block:** The master starter engine. This wakes up our dancer the millisecond the main green flag is touched.
2. **Hop Block (Set parameter value to 2):** Tells our dancer to bend his knees and bounce up into the air two grid units high!
3. **Grow Block (Set parameter value to 2):** Makes him dynamically expand in scale, simulating a powerful dance jump forward toward the screen.
4. **Shrink Block (Set parameter value to 2):** Resets his size parameter back to normal so he stays perfectly proportional on the stage.
5. **Forever Loop Arrow:** Wrap this bright orange arrow all the way around the hop and size blocks. This forces him to loop his energetic routine over and over again until you click stop!

Coding Tip: Make sure the Grow block and Shrink block use the exact same number parameter. If they don't, your dancer will keep growing larger and larger until he fills up the entire screen like a giant monster! 🦖

8. Assembling the Code Train for Garba

Now, select your ****Garba Performer**** sprite icon. Her routine needs to look fluid, circular, and rhythmic. We will combine sideways motion blocks with full rotational commands.

THE GARBA CODE RECIPE GUIDE

Snap these blocks together in her workspace channel area from left to right:

Block Order	Visual Block Type	The Secret Choreography Instructions
Step 01		Starts her performance in perfect timing with her partner.
Step 02		Teaches her to slide gracefully across the stage floor three steps.
Step 03		Triggers a sharp audio snap that mimics her clapping to the music.
Step 04		Commands her to spin all the way around on her tiptoes.
Step 05		Locks the entire line in a continuous loop so she never stops her graceful dance.

LIVE LAB INSPECTION SCRIPT

Teacher: "Press that big Green Flag at the top of your screen, software engineers! Look at your stage! We have a high-energy hopper on the left and a beautiful twirling spinner on the right. You just coded a cross-cultural digital festival!" ✨🌟

9. Becoming a Professional Dance Doctor!

Sometimes, even the best choreographers make a mistake, and their dancers do funny, broken things on stage. In computer science, a code mistake is called a **Bug** , and finding it is called **Debugging**. Let's look at two common dance bugs and learn how to squash them:

BUG CASE #1: THE GIANT GROWING PERFORMER

The Sad Sight: You click the green flag, and your Bhangra dancer starts hopping, but every time he lands, he grows bigger and bigger until his giant head blocks out the entire background scenery!

The Easy Fix: Check your parameters! This bug happens when you add a purple  **Grow** block but forget to add the matching  **Shrink** block right after it. Remember, what goes up must come down, and what grows big must shrink back to normal size!

BUG CASE #2: THE DIZZY LEANING PERFORMER

The Sad Sight: Your Garba dancer starts spinning, but instead of finishing her turn upright, she stops halfway through and stays tilted sideways like she is falling over! 

The Easy Fix: In Scratch Jr, it takes a numerical parameter value of ****12**** to make a perfect, complete circular turn. If you typed in a small number like 3 or 6, she will stop halfway through and get stuck tilted. Change that number to 12 to make her stand tall and straight!

10. The Dance Showcase Trivia Game!

Let's check your super-coder brain power! We have two tricky mystery questions to solve based on our dance project today. Read the descriptions carefully and choose the correct answer card.

QUESTION 1: THE FAST TEMPO MIX-UP

The Problem: Your dance teacher tells you the music is speeding up fast! You need your Scratch Jr character to hop up and down much quicker. What part of the block should you adjust to change its speed?

- **Option A: Swap the backdrop picture.** 

✗ **INCORRECT.** The background only changes the scenery; it has no control over character movement speeds.

- **Option B: Change the number parameter value on the blue Hop block.** 

✓ **CORRECT ANSWER!** Lowering or raising the numerical parameter numbers changes the height or pace of the character's movement script!

- **Option C: Change the color of the character's dress.** 

✗ **INCORRECT.** Paint colors look beautiful, but they don't change how fast a script runs.

QUESTION 2: THE ONE-TIME DANCER CHALLENGE

The Problem: You want your dancer to perform her routine just ****one single time**** and then stop and bow, instead of dancing forever. What block should you remove from your code train?

- **Option A: Remove the orange Forever Loop block.** 

✓ **CORRECT ANSWER!** The orange loop forces blocks to repeat endlessly. Removing it lets the sequence play once from left to right and then stop peacefully.

- **Option B: Remove the yellow Green Flag block.** 

✗ **INCORRECT.** If you remove the flag block, your engine is gone and the dancer will never move at all!

11. Your Take-Home Code Mission: Family Grooves!

Congratulations, Master Digital Choreographers! You have officially graduated from our Scratch Jr Dance Party Lab. To celebrate your new coding skills, you are being handed an official take-home family mission assignment.



THE FAMILY DANCE RECORDING TASK

When you get home tonight, open Scratch Jr on your tablet and show your project to your parents and siblings. Ask an adult family member to teach you a traditional dance move from your family's heritage (like an elegant classical step, a wedding slide, or a festive folk dance). Then, try to recreate those rhythmic movements on your tablet using your Scratch Jr block grid!

MY PERSONAL DANCE CODING LOG SHEET

1. Name of My Family Heritage Dance Style:

2. What blocks did I use for this new dance style? (Check the boxes below):

☐

Green Flag

☐

Move Left/Right

☐

Turn Spin

☐

Forever Loop

3. Did my parents test and play my dance game? ☐ YES! ☐ NOT YET!

Planrs Academy Curriculum Certification: This integrated lesson map connects physical coordination with early computational logic, aligning perfectly with K-2 primary developmental milestones.